2013 F=ma Exam Review

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Problem 3

A is true because $\sin 2\theta \leq 1$ and $R = v^2 \sin 2\theta / g$, so $gR \leq v^2$. If v increases with R held fixed, the min value of t decreases because you can shoot at a smaller angle faster, and the max value of t increases because you can shoot at a larger angle. Overall, increasing v expands the range of time values. If R increases with v held fixed, the min value of t increases because you need to shoot at a higher angle, and the max value of t decreases because you need to shoot at a less steep angle. Overall, increasing R narrows the range of time values.

Problem 4

Balance torques about the hinge.

Problem 9

Yes, the initial $p = Ft$, but that's not an option.

$$x = v_{\text{avg}} t, \qquad t = \frac{x}{v_{\text{avg}}} = \frac{x}{v/2} = \frac{2x}{v}, \qquad Ft = \frac{2Fx}{v}.$$

Or: $K_i = \frac{1}{2}mv^2 = \frac{1}{2}pv$, so $p = 2K_i/v$, and from the previous problem $K_i = Fx$, so $p = 2Fx/v$.

Problem 12

Don't skip writing out the force equations for this one. The frictionless fluid contributes to the mass but not to the rotational inertia about the com, because it's not rotating.

$2Mg \sin\theta - f = 2Ma$, $\quad fR = I\alpha$, $\quad I = \frac{2}{3}MR^2$ (only the M of the shell rotates), $\quad \alpha = \dfrac{a}{R}$

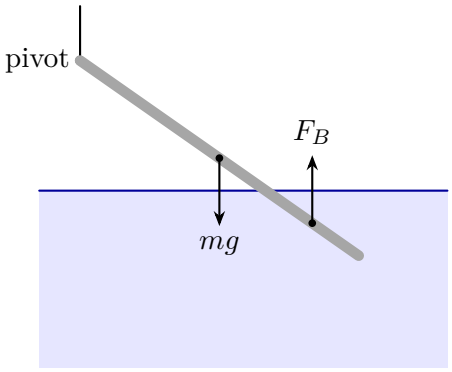
(roll without slip), so $f = \frac{2}{3}Ma$. Proceed to solve for a.

Problem 13

For the rings to be part of Saturn, $v = r\omega$ (they have to rotate at the same ω). Don't mix up the ω of the planet's rotation and the ω of an orbit: $v^2 \propto 1/R$ is the result for orbiting satellites, not for something that's part of the rigid rotating body. For the rings to be part of Saturn, $v \propto R$, that's it; the ω of the planet's rotation is just some constant.

Problem 15

Torque balance. Set a θ; it'll cancel. The buoyant force acts at the midpoint of the portion of the rod under water. $V_{\text{sub}}/V = L_{\text{sub}}/L$ (the cross-sectional areas cancel).



Problem 18

Closed system $\rightarrow \Delta p = 0$, v_{cm} is constant $= 0$, so $v = 0$, so $\Delta x_{cm} = 0$. ΔL is also conserved. Remember: L about a point is mv times the lever arm.

Problems 23 & 24

Set up the simple energy and force equations, and solve. The answer isn't automatically in the same format, hinting that you need the relation between h and H.

Problem 25

The first sentence lets you find an expression for μ. Read carefully: P is a force, not power, in this problem. Because the force is horizontal, there is a component perpendicular to the incline, so you must account for greater friction: $\mu(W\cos\theta + P\sin\theta)$. Carry out the algebra to solve for P in terms of W.